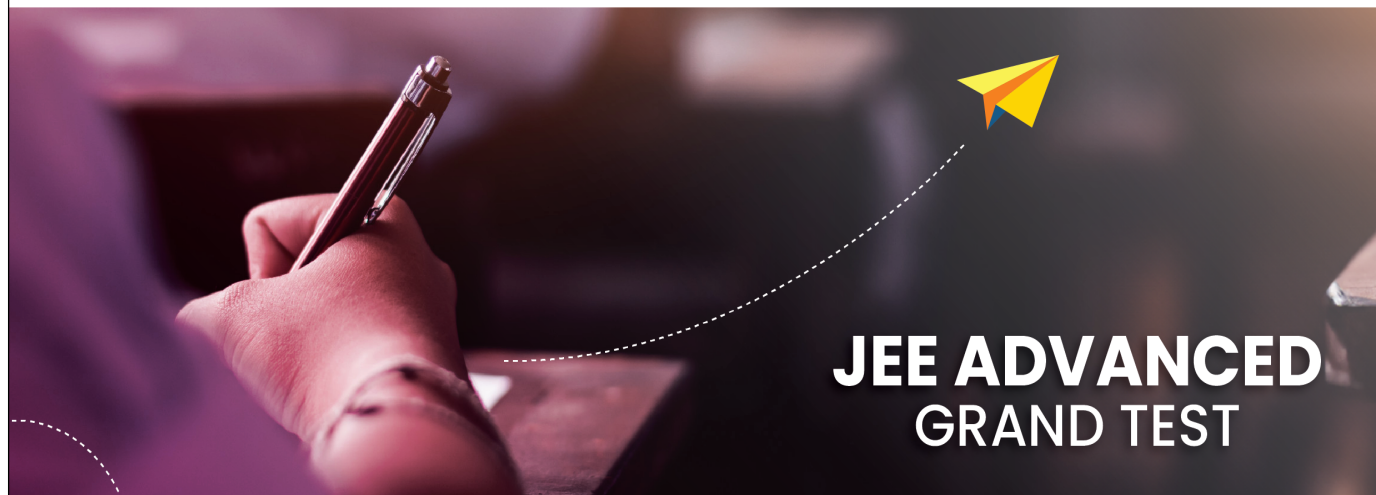




**Sri Chaitanya**  
Educational Institutions



## JEE ADVANCED GRAND TEST



**Sri Chaitanya** IIT Academy.,India.

✦ A.P ✦ T.S ✦ KARNATAKA ✦ TAMILNADU ✦ MAHARASTRA ✦ DELHI ✦ RANCHI

*A right Choice for the Real Aspirant*

ICON Central Office - Madhapur - Hyderabad

Sec: **Sr.Super60\_Elite, Target & LIIT-BTs**

**Paper -2(Adv-2023-P2-Model)**

**Date: 08-05-2024**

**Time: 02.30Pm to 05.30Pm**

**GTA-31**

**Max. Marks: 180**

08-05-2024\_**Sr.Super60\_Elite, Target & LIIT-BTs**\_Jee-Adv(2023-P2)\_GTA-31\_Syllabus

**MATHEMATICS : TOTAL SYLLABUS**

**PHYSICS : TOTAL SYLLABUS**

**CHEMISTRY : TOTAL SYLLABUS**

**Name of the Student:** \_\_\_\_\_

**H.T. NO:**

|  |  |  |  |  |  |
|--|--|--|--|--|--|
|  |  |  |  |  |  |
|--|--|--|--|--|--|

**JEE-ADVANCE-2023-P2-Model**

Time:3Hr's

**IMPORTANT INSTRUCTIONS**

Max Marks: 180

**MATHEMATICS:**

| Section                 | Question Type                                            | +Ve Marks | - Ve Marks | No.of Qs  | Total marks |
|-------------------------|----------------------------------------------------------|-----------|------------|-----------|-------------|
| Sec – I(Q.N : 1 – 4)    | Questions with Single Correct Choice                     | +3        | -1         | 4         | 12          |
| Sec – II(Q.N : 5 – 7)   | Questions with Multiple Correct Choice with partial mark | +4        | -2         | 3         | 12          |
| Sec – III(Q.N : 8 – 13) | Questions with Integer Answer Type                       | +4        | 0          | 6         | 24          |
| Sec – IV(Q.N : 14 – 17) | Paragraph NUMERICAL VALUE                                | 3         | 0          | 4         | 12          |
| <b>Total</b>            |                                                          |           |            | <b>17</b> | <b>60</b>   |

**PHYSICS:**

| Section                  | Question Type                                            | +Ve Marks | - Ve Marks | No.of Qs  | Total marks |
|--------------------------|----------------------------------------------------------|-----------|------------|-----------|-------------|
| Sec – I(Q.N : 18 – 21)   | Questions with Single Correct Choice                     | +3        | -1         | 4         | 12          |
| Sec – II(Q.N : 22 – 24)  | Questions with Multiple Correct Choice with partial mark | +4        | -2         | 3         | 12          |
| Sec – III(Q.N : 25 – 30) | Questions with Integer Answer Type                       | +4        | 0          | 6         | 24          |
| Sec – IV(Q.N : 31 – 34)  | Paragraph NUMERICAL VALUE                                | 3         | 0          | 4         | 12          |
| <b>Total</b>             |                                                          |           |            | <b>17</b> | <b>60</b>   |

**CHEMISTRY:**

| Section                  | Question Type                                            | +Ve Marks | - Ve Marks | No.of Qs  | Total marks |
|--------------------------|----------------------------------------------------------|-----------|------------|-----------|-------------|
| Sec – I(Q.N : 35 – 37)   | Questions with Single Correct Choice                     | +3        | -1         | 4         | 12          |
| Sec – II(Q.N : 38 – 41)  | Questions with Multiple Correct Choice with partial mark | +4        | -2         | 3         | 12          |
| Sec – III(Q.N : 42 – 47) | Questions with Integer Answer Type                       | +4        | 0          | 6         | 24          |
| Sec – IV(Q.N : 48 – 51)  | Paragraph NUMERICAL VALUE                                | 3         | 0          | 4         | 12          |
| <b>Total</b>             |                                                          |           |            | <b>17</b> | <b>60</b>   |

Sec: Sr.Super60\_Elite, Target &amp; LIIT-BTs

Space for rough work

Page 2

**Sri Chaitanya**  
Educational Institutions**THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1  
IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023**

**JEE MAIN  
2023**  
SINGARAJ  
VENKAT KOUNDINYA  
Sri Chaitanya  
100th Class  
**300**  
**300**

**1**

**JEE Advanced  
2023**  
VARSHIL  
CHIDVILAS REDDY  
Sri Chaitanya  
100th Class  
**341**  
**360**

**1**

**NEET  
2023**  
BORA VARUN  
CHAKRAVARTHI  
Sri Chaitanya  
100th Class  
**720**  
**720**

**1**

**MATHEMATICS****Max Marks: 60****SECTION-I  
(SINGLE CORRECT ANSWER TYPE)**This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

**Full Marks** : +3 If **ONLY** the correct option is chosen;**Zero Marks** : 0 If none of the options is chosen (i.e. the question is unanswered);**Negative Marks** : -1 In all other cases

- Let  $f(x) = e^{\{e^{|x|} \operatorname{sgn} x\}}$  and  $g(x) = e^{\lfloor e^{|x|} \operatorname{sgn} x \rfloor}$ ,  $x \in \mathbb{R}$  where  $\{x\}$  and  $\lfloor \cdot \rfloor$  denotes the fractional part and integral part functions respectively. Also  $h(x) = \ln(f(x)g(x))$  then for all real  $x$ ,  $h(x)$  is  
 A) an odd Function  
 B) an even function  
 C) neither an odd nor an even function  
 D) both odd as well as even function
- Suppose  $a, b, c$  are in A.P. and  $a^2, b^2, c^2$  are in G.P. If  $a < b < c$  and  $a + b + c = \frac{3}{2}$ , then the value of  $c$  is  
 A)  $\frac{1}{2\sqrt{2}}$       B)  $\frac{\sqrt{3}}{2}$       C)  $\frac{1}{2} + \frac{1}{\sqrt{3}}$       D)  $\frac{1}{2} + \frac{1}{\sqrt{2}}$
- Find the coefficient of  $x^5$  in the expansion of  $(3x^2 - x + 2)^6$   
 A) -5049      B) 5051      C) -5052      D) 5050
- The largest two digit prime factor of  ${}^{200}C_{100}$  is  
 A) 61      B) 59      C) 17      D) none of these

**SECTION – II  
(ONE OR MORE CORRECT ANSWER TYPE)**This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

**Full Marks** : +4 **ONLY** if (all) the correct option(s) is(are) chosen;**Partial Marks** : +3 If all the four options are correct but **ONLY** three options are chosen;**Partial Marks** : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;**Partial Marks** : +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;**Zero Marks** : 0 If none of the options is chosen (i.e. the question is unanswered);**Negative Marks** : -2 In all other cases.

For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to correct answers, then choosing **ONLY** (A), (B) and (D) will get +4 marks; choosing **ONLY** (A) and (B) will get +2 marks; choosing **ONLY** (A) and (D) will get +2 marks; choosing **ONLY** (B) and (D) will get +2 marks; choosing **ONLY** (A) will get +1 mark; choosing **ONLY** (B) will get +1 mark; choosing **ONLY** (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

Sec: **Sr.Super60\_Elite, Target & LIIT-BTs**

Space for rough work

Page 3

**Sri Chaitanya**  
Educational Institutions**THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1  
IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023**
**JEE MAIN  
2023**  
SINGARAU  
VENKAT KOUNDURU  
Sri Chaitanya  
Elite 2nd Class  
**300**  
**300**
**RANK  
1**
**JEE Advanced  
2023**  
SIVAVILAS  
CHIDVILAS REDDY  
Sri Chaitanya  
Elite 2nd Class  
**341**  
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**RANK  
1**
**NEET  
2023**  
SODA VARUN  
CHAKRAVARTY  
Sri Chaitanya  
Elite 2nd Class  
**720**  
**720**
**RANK  
1**





5. Maximum number of regions into which 10 lines can divide a plane, is  
 A) 42                      B) 56                      C)  ${}^8P_2$                       D)  ${}^7P_2$
6. Let the complex number  $z = x + iy$  (where  $x, y \in R$ ) satisfy  $\cot^{-1}(\log_3|2z+1|) > \cot^{-1}(\log_3|2z-1|)$ , then x can be  
 A) -1                      B) -2                      C) 1                      D) 2
7. Which of the following when simplified reduces to unity?  
 A)  $\log_3 \log_{27} \log_4 64$                       B)  $2\log_{18}(\sqrt{2} + \sqrt{8})$   
 C)  $\log_2 \sqrt{10} + \log_2 \left(\frac{2}{\sqrt{5}}\right)$                       D)  $-\log_{\sqrt{2}-1}(\sqrt{2} + 1)$

### SECTION-III (NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:  
**Full Marks: +4 if ONLY the correct integer is entered; Zero Marks : 0 In all other cases.**

8. Let  $f(x) = ax^4 + bx^2 + 3x + 7$  and  $f(-4) = 2286$  and  $f(4) = N$ . If k be the number of ways in which the number N can be resolved as a product of two divisors which are relatively prime then k-7 is \_\_\_\_\_.
9. A set of numbers contains five values. The largest value is 500, and the range is 100. Consider the following statements about the set of numbers:  
 I) The largest possible value for the mean is 480.  
 II) The smallest possible value for the mean is 420.  
 III) The largest possible value for the median is 500.  
 IV) The smallest possible value for median is 400.  
 V) The smallest possible value for the standard deviation is 0.  
 The number of correct statements among the above is \_\_\_\_\_.
10. If the solution of the differential equation  $e^{\frac{x}{y}(1-y^2)} \left[ y \frac{dx}{dy} - x \right] + \left[ xy^2 + y^3 \frac{dx}{dy} \right] = 0$  is  $Ae^{xy} + Be^{\frac{x}{y}} + c = 0$ , then  $|A+B|$  is \_\_\_\_\_.
11. Let  $S_1$  be the locus of the point 'P' which moves such that  $OP=1$ . The plane  $x+y+z=1$  cuts the curve  $S_1$  in another curve  $S_2$ . The volume of solid formed by joining the point O with every point on the curve  $S_2$ . (O is origin) is in the form  $V = \frac{a\pi\sqrt{c}}{b}$  where  $a, b, c \in N$  and a, c are prime, then  $a+b+c =$  \_\_\_\_\_.





12. Let  $A(t) = [a_{ij}]$  is a matrix of order  $3 \times 3$  given by  $a_{ij} = \begin{cases} 2 \cos t, & \text{if } i = j \\ 1 & \text{if } |i - j| = 1 \\ 0 & \text{otherwise} \end{cases}$  where  $|A(t)|$  is determinant value of matrix  $A(t)$ , then  $\lim_{t \rightarrow 0} |A(t)| |A(4t)|$  is equal to
13. A point  $P\left(\frac{e^t + e^{-t}}{2}, \frac{e^t - e^{-t}}{2}\right)$  traces a locus  $S=0$  in X-Y plane (t being a non negative parameter). A fixed point  $P_0$ , having parameter  $t_0$ , lies on the curve  $S=0$ . The area bounded by the curve, line  $OP_0$  and x-axis is 240sq. units (O being the origin), find the value of  $t_0$ .

### SECTION-IV (PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- **Full Marks: +3** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks: 0** in all other cases

#### PARAGRAPH-I

There is a point inside an equilateral triangle ABC of side d whose distance from the vertices is 3, 4, 5. Rotate the triangle and P through  $60^\circ$  about C. Let A go to  $A'$  and P to  $P'$ .

14. The area of triangle  $PAP'$  is

15. The value of  $d^2$  is

#### PARAGRAPH-II

Let  $f(x) = 1 + \int_0^1 (xe^y + ye^x) f(y) dy$  when x and y are independent variables.

16. If complete solution set of 'x' for which function  $h(x) = f(x) + 3x$  is strictly increasing  $(-\infty, k)$  then  $\left[\frac{4}{3}e^k\right]$  equals to: (where  $[.]$  denotes greatest integer function):
17. If acute angle of intersection of the curve  $\frac{x}{2} + \frac{y}{3} + \frac{1}{3} = 0$  and  $y = f(x)$  be  $\theta$  then  $\tan \theta$  equals to:





## PHYSICS

Max Marks: 60

SECTION-I  
(SINGLE CORRECT ANSWER TYPE)This section contains **FOUR (04)** questions.

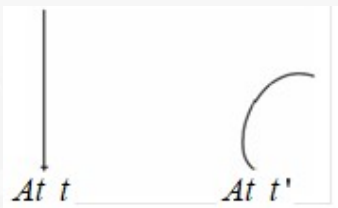
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -1 In all other cases

18. The figure shows a plane wave-front film at a time  $t$  and at time  $t'$  respectively. In the time interval  $(t' - t)$  the wave may have passed through



- A) A prism and a concave lens      B) A prism and a convex lens  
C) A convex lens and a concave lens      D) A plane mirror and concave lens
19. A positron is an antiparticle of electron. A photon is a particle of light, which has no charge but non-zero energy, momentum proportional to the light frequency. If an electron and positron moving towards each other with equal and opposite velocities. Then they
- A) Can annihilate into one photon conserving both energy and momentum  
B) Can annihilate into one photon without conserving both energy and momentum  
C) Cannot annihilate into one photon because energy cannot be conserved  
D) Cannot annihilate into one photon because momentum cannot be conserved
20. A radioactive material with decay constant  $\lambda$  is produced at a rate of  $P$  nuclei per second. Find the number of nuclei  $N$  present 't' second after the production starts

A)  $N = \frac{P(1 - e^{-\lambda t})}{2\lambda}$

B)  $N = \frac{2P(1 - e^{-\lambda t})}{\lambda}$

C)  $N = \frac{P(1 - e^{-\lambda t})}{\lambda}$

D)  $N = \frac{P(1 - 2e^{-\lambda t})}{\lambda}$

JEE MAIN  
2023SINGARAO  
VENKAT KOUNDURU  
Sri Chaitanya  
The 2nd Class300  
300

RANK

1

JEE Advanced  
2023VARUN  
CHIVILAS REDDY  
Sri Chaitanya  
The 2nd Class341  
360

RANK

1

NEET  
2023SODA VARUN  
CHAKRAVARTY  
Sri Chaitanya  
The 2nd Class720  
720

RANK

1





21. A cylindrical glass rod of radius 10 cm and refractive index  $\sqrt{3}$  rests horizontally on a horizontal surface. A horizontal ray of light moving perpendicular to the axis of the rod is incident on it. At what height from the surface should the ray be incident so that it leaves the rod at a height of 10 cm above the surface.

A) 16.24 cm      B) 18.66 cm      C) 14.14 cm      D) 12.24 cm

**SECTION – II**  
**(ONE OR MORE CORRECT ANSWER TYPE)**

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
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Full Marks : +4 **ONLY** if (all) the correct option(s) is(are) chosen;

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen;

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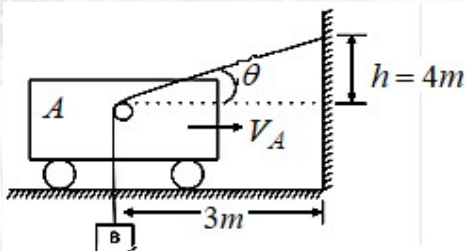
Partial Marks : +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -2 In all other cases.

For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

22. The string shown in the figure is passing over small smooth pulley rigidly attached to trolley A. The trolley starts moving with constant speed equal to  $V_A$ , then vertical speed and magnitude of acceleration of block B at the initial instant shown in figure is



A)  $a_B = \frac{16V_A^2}{125}$       B)  $a_B = 0$       C)  $V_B = \frac{3}{5}V_A$       D)  $V_B = V_A$

23. When a lithium is bombarded by deuteron, a neutron is observed to emerge and  $Be^8$  is formed. Relevant masses in amu

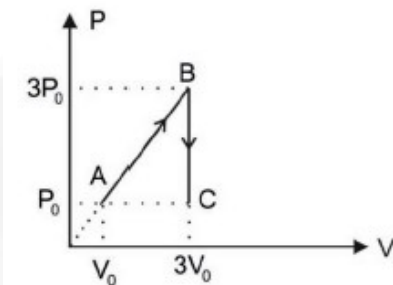
are  $[{}_0n^1 = 1.00893, {}_3Li^7 = 7.01784, {}_1H^2 = 2.01472, {}_4Be^8 = 8.00776]$ . then

- A) The 'Q'-value of the reaction is 14.783 MeV  
B) The 'Q'-value of the reaction is 12.775 MeV  
C) Mass defect in the nuclear reaction is 0.01587 amu  
D) Mass defect in the nuclear reaction is 0.0642 amu





24. 2 moles of a diatomic gas undergoes the process as illustrated in the PV graph from A to B, and then from B to C



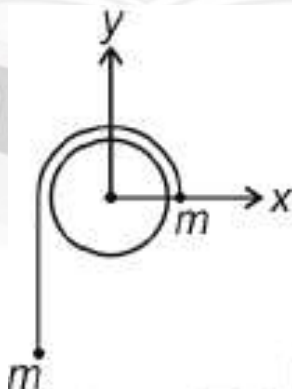
- A) Process AB is  $PV^{-1} = \text{constant}$
- B) The change in temperature in the process from A to B is  $\frac{4P_0V_0}{R}$
- C) Heat taken by gas in the process A to B is  $24P_0V_0$
- D) Total work done by the gas in the process  $A \rightarrow B \rightarrow C$  is  $4P_0V_0$

### SECTION-III (NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**.
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- Answer to each question will be evaluated according to the following marking scheme:  
Full Marks: +4 If **ONLY** the correct integer is entered; Zero Marks : 0 In all other cases..

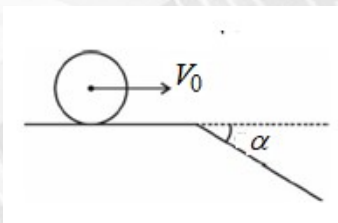
25. Figure shows a smooth cylindrical pulley of radius  $R$  with centre at origin of co-ordinates. An ideal thread is thrown over it on the two ends of ideal thread two identical masses are tied initially at rest with coordinates  $(R, 0)$  and  $(-R, -R)$  respectively. If mass at x-axis is given a slight upward jerk, it leaves contact with pulley at  $(R \cos \theta, R \sin \theta)$ .

Then  $\frac{\theta}{\sin \theta}$  is equal to

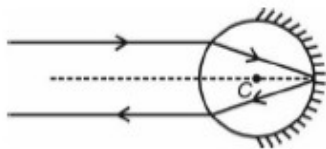




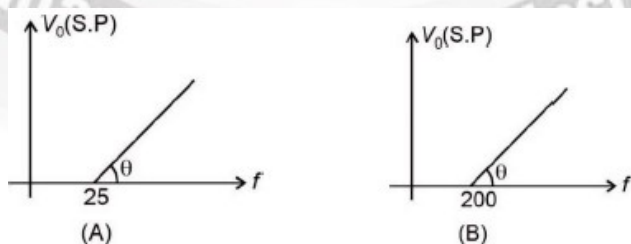
26. The potential energy of a particle in a certain field has the form  $U = \frac{a}{r^2} - \frac{b}{r}$ , where  $a$  and  $b$  are positive constants,  $r$  is the distance from centre of the field. At  $r = \frac{N_1 a}{b}$ , particle is in equilibrium. Magnitude of maximum force of attraction is  $\frac{b^3}{N_2 a^2}$ . Find the product of  $N_1, N_2$
27. uniform circular disc of radius  $R$  rolls with velocity  $v_0$  on a horizontal plane changing into an inclined plane forming angle  $\alpha$  with the horizontal. The maximum value of velocity  $v_0$  which still permits the disc to roll on the inclined plane section without a jump is  $v_0 = \sqrt{\left(\frac{A}{B} \cos \alpha - \frac{C}{3}\right) g R}$ . Find the product of  $A, B, C$ .



28. A transparent cylinder has its right half polished so as to act as a mirror. A paraxial light ray incident from left, parallel to principal axis, exits parallel to that incident ray as shown. The refractive index of the material of cylinder is

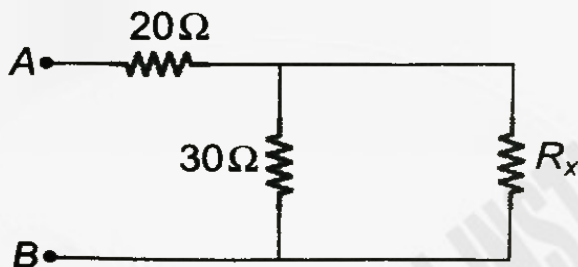


29. Radiation of frequency  $f$  falls on the surface of two metals A and B. Stopping potential vs frequency  $f$  graph for these two metals A and B is shown below



Find ratio  $\frac{\phi_B (\text{work function of } B)}{\phi_A (\text{work function of } A)}$

30. A circuit shown in the figure has resistance  $20\Omega$  and  $30\Omega$ . At what value of resistance  $R_x$  ( $\text{in}\Omega$ ) will the thermal power generated in it be practically independent of small variation of that resistance? The voltage between A and B is supposed to be constant in this case.

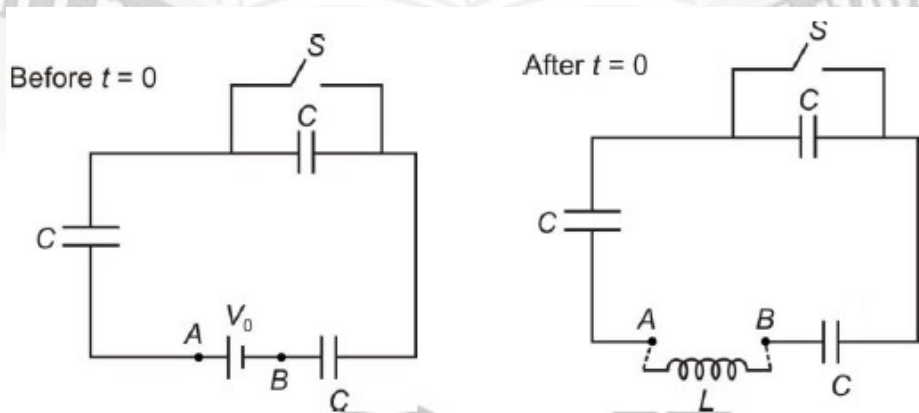


#### SECTION-IV (PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- Full Marks: +3** If ONLY the correct numerical value is entered at the designated place;
- Zero Marks:0** in all other cases

#### PARAGRAPH-I

Three capacitors each of capacitance  $C$  are connected in series as shown. Initially the switch  $S$  is open. Now capacitors are charged by a battery of emf  $V_0$  by connecting across A and B. At  $t = 0$  battery is disconnected and inductor of inductance  $L$  is connected between A and B. After then, switch  $S$  is closed at any further time  $t_0$ . Based on all these informations, now answer the following questions.

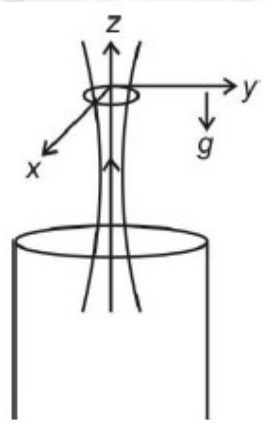




31. If  $t_0 = 2\pi\sqrt{\frac{LC}{3}}$ , Energy loss in the system is  $\frac{CV_0^2}{x}$ . Then  $x$  is \_\_\_\_\_.
32. If  $t_0 = \frac{\pi}{3}\sqrt{\frac{LC}{3}}$ ,  $Q_{\max}$  be charge amplitude and  $I_{\max}$  be current amplitude, then
- $$Q_{\max} = \frac{CV_0}{x}\sqrt{\frac{11}{2}}, x = \text{_____}.$$

**PARAGRAPH-II**

A thin superconducting ring is held above a vertical long solenoid as shown in the figure. Axis of solenoid is same as that of a ring. The cylindrically symmetric magnetic field around the ring in terms of vertical and radial component, is  $B_z = B_0(1 - \alpha z)$  and  $B_r = B_0 Br$  where  $B_0$ ,  $\alpha$  and  $B$  are positive constant and  $z$  and  $r$  are vertical and radial components respectively. Initially plane of ring is horizontal with no current flowing through it. Mass of the ring is  $m$ , radius  $r_0$  and self inductance  $L$ . If initially, coordinates of ring are  $(0,0,0)$  and it is released then



33. When the ring is at vertical position  $z$ , then current in the ring is  $\frac{B_0\alpha z\pi r_0^2}{xL}$ . Then  $x$  is.
34. Regarding forces acting on the ring due to various components of magnetic field tension in the ring is maximum when  $z = \frac{1}{x\alpha}$ . Then  $x$  is \_\_\_\_\_





## CHEMISTRY

Max Marks: 60

SECTION-I  
(SINGLE CORRECT ANSWER TYPE)This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;

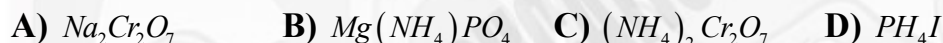
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -1 In all other cases

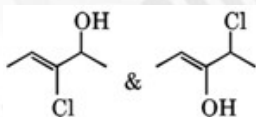
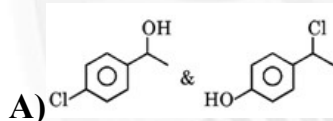
35. Which of the following reactions is an example of a redox reaction?



36. The coloured salt P was warmed with NaOH solution and colourless gas Q evolved and a coloured solution R was produced. The colour of salt P was different from that of solution R. On adding  $BaCl_2$  to R, a coloured PPT was obtained. Then the correct formula for salt P is

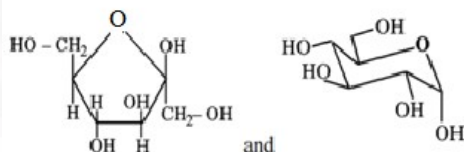


37. Select CORRECT option.



B) can be differentiated by Beilstein test

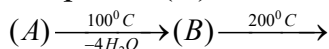
C) Monomer of polyethene is Vinyl chloride



D) can be distinguished by Seliwanoff's test

38. Chlorine on reaction with iodate ion in basic medium, according to the given equation :  
 $IO_3^- + xOH^- + Cl_2 \rightarrow (A) + yH_2O + zCl^-$  [x, y, z are stoichiometric coefficients in balance equations]

Compound (A) on heating in acidic medium undergoes,



Identify the compound (C)?





## SECTION – II

### (ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

**Full Marks : +4 ONLY** if (all) the correct option(s) is(are) chosen;

**Partial Marks : +3** If all the four options are correct but **ONLY** three options are chosen;

**Partial Marks : +2** If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;

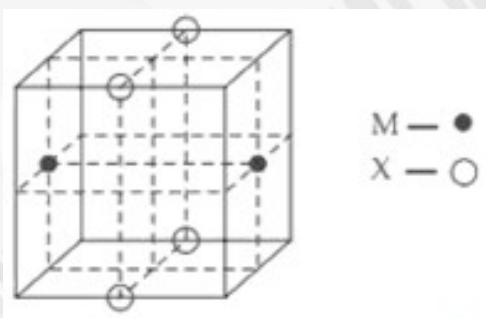
**Partial Marks : +1** If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;

**Zero Marks : 0** If none of the options is chosen (i.e. the question is unanswered);

**Negative Marks : -2** In all other cases.

For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to correct answers, then choosing **ONLY** (A), (B) and (D) will get +4 marks; choosing **ONLY** (A) and (B) will get +2 marks; choosing **ONLY** (A) and (D) will get +2 marks; choosing **ONLY** (B) and (D) will get +2 marks; choosing **ONLY** (A) will get +1 mark; choosing **ONLY** (B) will get +1 mark; choosing **ONLY** (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

39. The cubic unit cell structure of a compound containing cation M and anion X is shown below. When compared to the anion, the cation has smaller ionic radius. Choose the incorrect statement (s).



- A) The empirical formula of the compound is  $MX_2$
- B) The cation M and anion X have different coordination geometries
- C) The ratio of M-X bond length to the cubic unit cell edge length is 0.866
- D) The ration of the ionic of cation M to anion X is 0.414
40. When 5 mL, of a 1.0M HCl solution is mixed with 5 mL of a 0.1M NaOH solution temperature of solution increases by  $t^\circ\text{C}$ . which of the following can be predicted from this observation
- A) If 5 mL of 0.1M HCl is mixed with 5 mL 0.1 M,  $\text{NH}_3$  solution, the temperature rise will be less than  $t^\circ\text{C}$
- B) If 5 mL of 1.0M  $\text{CH}_3\text{COOH}$  is mixed with 5 mL of 0.1M NaOH, the temperature rise will be less than  $t^\circ\text{C}$
- C) If 10 mL of 1.0M HCl is mixed with 10 mL of 0.1 M NaOH, the temperature rise will be  $2t^\circ\text{C}$
- D) If 10 mL of 1.0M HCl is mixed with 10 mL of 0.1 M NaCl, the temperature rise will be  $2t^\circ\text{C}$



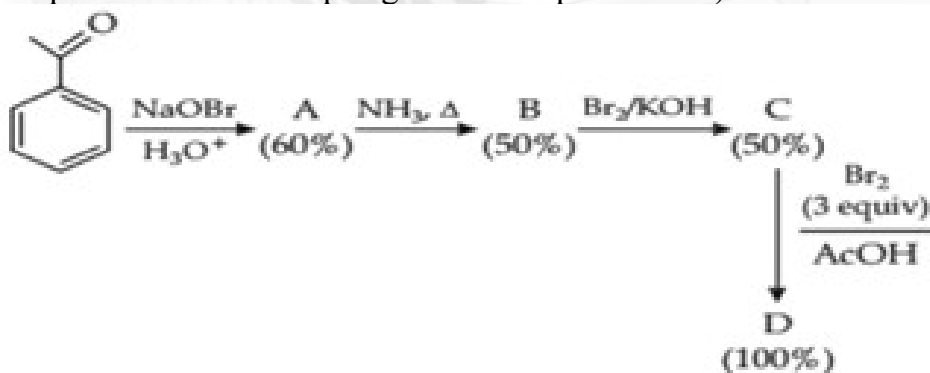


41. The following statements related to the adsorption of gases on a solid surface. Identify the correct statement among them:
- Enthalpy of adsorption is negative
  - On adsorption, the residual forces on surface are increased.
  - On adsorption, decrease in surface energy appears as heat.
  - Entropy of adsorption is negative.

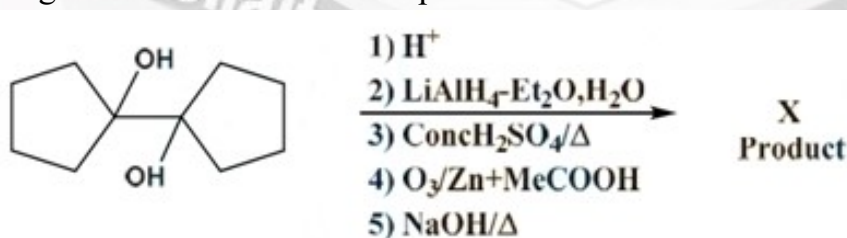
**SECTION-III**  
**(NON-NEGATIVE INTEGER)**

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER..**
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:  
**Full Marks: +4 If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.**

42. In 1L saturated solution of  $\text{AgCl}$  [ $K_{sp}(\text{AgCl}) = 1.6 \times 10^{-10}$ ] 0.1 mol of  $\text{CuCl}$  [ $K_{sp}(\text{CuCl}) = 1.0 \times 10^{-6}$ ] is added. The resultant concentration of  $\text{Ag}^+$  in the solution is  $1.6 \times 10^{-x}$ . The value of 'x' is
43. In the following reaction sequence, the amount of D [in (g)] formed from 10 moles of acetophenone is .....  
(Atomic weights in  $\text{g mol}^{-1}$ : H=1, C=12, N=14, O=16, Br=80. The yield (%) corresponding to the production each step is given in the parenthesis)



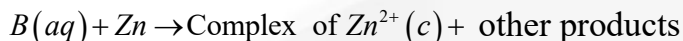
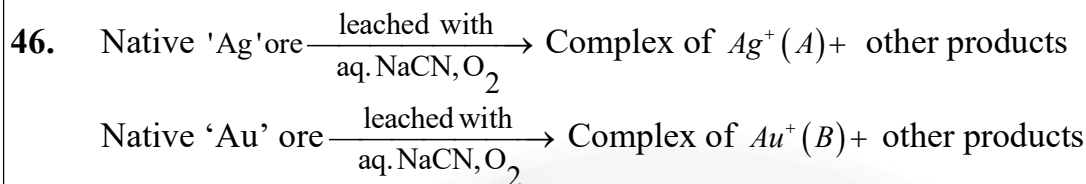
44. Degree of unsaturation of the product 'X' is



45. Find the number of the paramagnetic and inner orbital complex in the following :
- $[\text{Ni}(\text{CO})_4]$ ,  $[\text{Ni}(\text{CN})_4]^{2-}$ ,  $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$ ,  $[\text{Cu}(\text{NH}_3)_4]^{2+}$ ,  $[\text{Cr}(\text{NH}_3)_6]^{3+}$ ,  $[\text{Fe}(\text{CN})_6]^{3-}$







Coordination numbers of metal ions complexes A, B, C are p, q, r respectively. The spin only magnetic moment of metal ion complex 'C' in B. M. is 's'. The value of  $(p+q+r-s)$  is

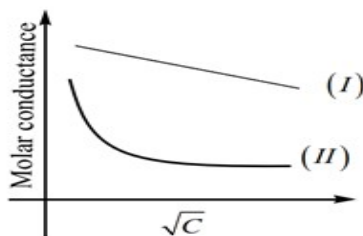
47. Compound A on treatment with  $\text{NaIO}_3$  in  $\text{HCl}$ ; gives a violet coloured (B). The violet colour disappears on treatment with (C), producing A and D. Compound 'C' is formed by reaction between sulphur and sodium sulphite. Find the maximum number of bonds between identical atoms having equal bond length in compound D. Find the Maximum number of bonds between identical atoms of different oxidation states having equal bond length in compound D.

### SECTION-IV (PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- Full Marks: +3** if ONLY the correct numerical value is entered at the designated place;
- Zero Marks: 0** in all other cases

#### PARAGRAPH-I

Molar conductance of electrolytes varies with  $\sqrt{c}$  generally as (I) or (II) as shown.



Answer the following questions based on above information

48. A saturated solution of silver bromide is made in  $10^{-7} M \text{ AgNO}_3$  solution

Given:  $K_{sp} \text{ of } \text{AgBr} = 3 \times 10^{-13}$

$$\lambda_m^\infty(\text{Ag}^+) = 6 \times 10^{-3} S \text{ m}^2 \text{ mol}^{-1}$$

$$\lambda_m^\infty(\text{NO}_3^-) = 7 \times 10^{-3} S \text{ m}^2 \text{ mol}^{-1}$$

$$\lambda_m^\infty(\text{Br}^-) = 8 \times 10^{-3} S \text{ m}^2 \text{ mol}^{-1}$$



For water,  $k = 7.5 \times 10^{-6} \text{ S m}^{-1}$

The conductivity of the solution is  $P \times 10^{-6} \text{ sm}^{-1}$ . Find the value of P.

49. The limiting molar conductivities of  $\text{NaI}$ ,  $\text{NaCH}_3\text{CO}_2$  and  $\text{Mg}(\text{CH}_3\text{CO}_2)_2$  are  $12.69 \text{ mS m}^2 \text{ mol}^{-1}$ ,  $9.10 \text{ mS m}^2 \text{ mol}^{-1}$  and  $18.78 \text{ mS m}^2 \text{ mol}^{-1}$  respectively (all at  $25^\circ\text{C}$ ). The limiting molar conductivity of  $\text{MgI}_2$  at this temperature is 'x'  $\text{S m}^2 \text{ mol}^{-1}$  (approx) then 'x' is

### PARAGRAPH-II

Entatiometric compounds are pair of compounds that are non-superimposable mirror image and rotate the plane of polarized light by exactly the same amount but in opposite direction. A solution of equal amount of two enantiomers is called a racemic mixture and is optically inactive. Sometimes we have mixtures that are neither optically pure (all are enantiomers) nor racemic (equal amount of two enantiomers). In these cases, we specify

the optical purity of the mixture. Optical purity (O.P) =  $\frac{\text{Observed notation}}{\text{rotation of pure enantiomer}} \times 100$

The enantiomer excess (e.e) is a similar method for expressing the relative amounts of enantiomers in a mixture.

$$\text{o.p} = \% \text{e.e} = \frac{\text{more of one enantiomer-moles of other enantiomer}}{\text{total moles of both enantiomers}} \times 100$$

50. Pure (-) enantiomer of an optically active compound (A) has a specific rotation  $[\alpha]_D^{25} = -50^\circ$ . Calculate the percentage of the (-) enantiomer present in a sample where specific rotation is  $-35^\circ$ .
51. What is the observed rotation when 0.1M solution of (R) -2-butanol is mixed with an equal volume of 0.05M solution of Racemic ( $\pm$ ) 2-butanol and the solution is kept in a polarimeter cell of 50 cm long? Specific rotation of (R) -2-butanol is  $14.06^\circ$ .







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